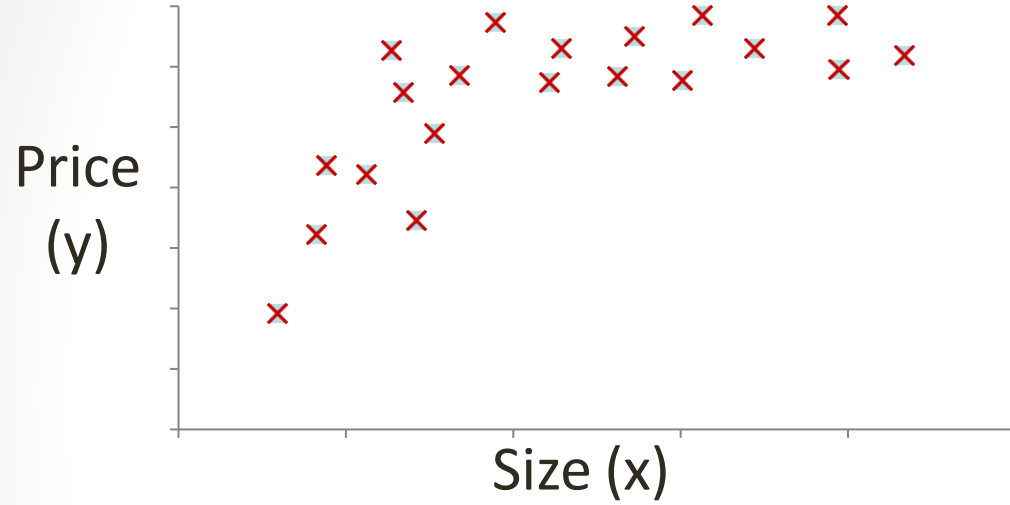


Machine Learning

Instructor: Hafiz Abdul Rehman

Lecture 12: Polynomial Regression

Linear Regression with one variable



Feature scaling becomes important as now:

Size = 1 – 1000

Size² = 1 – 1000,000

Size³ = 1 – 1000,000,000

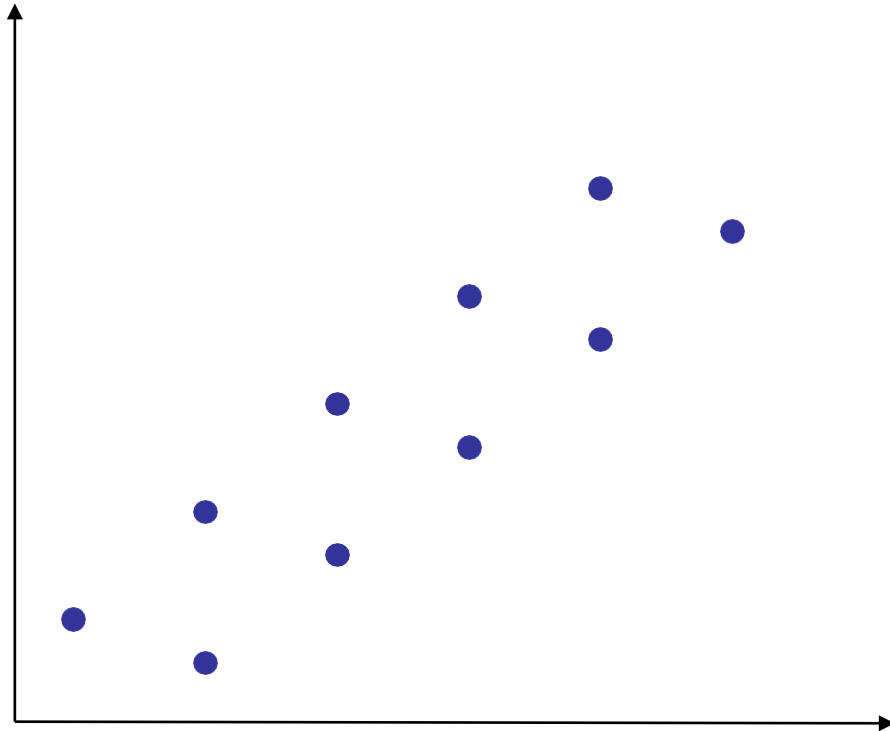
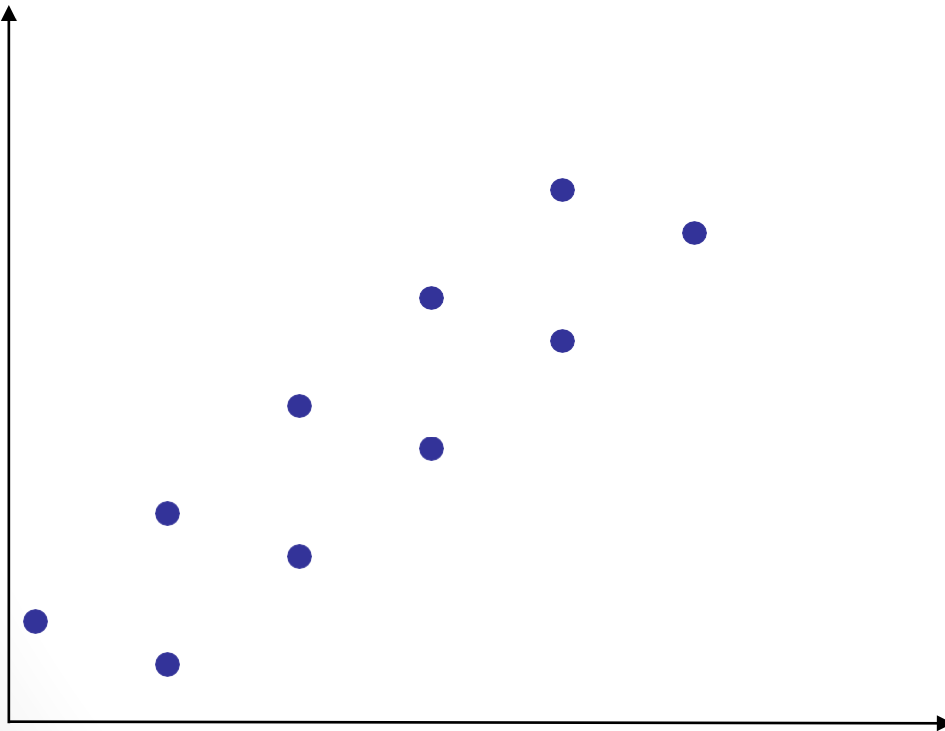
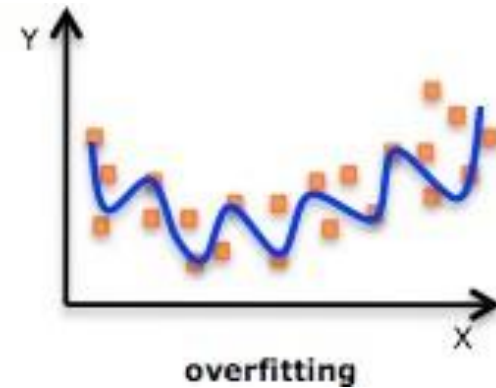
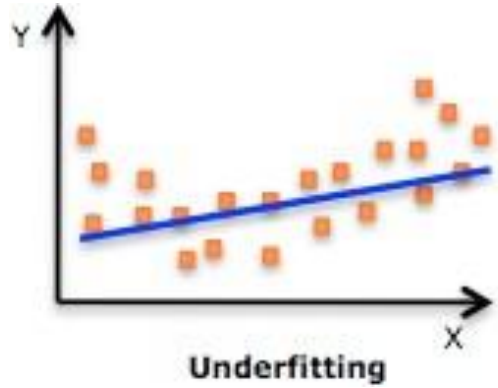
$$\begin{aligned}h_{\theta}(x) &= \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \theta_3 x_3 \\&= \theta_0 + \theta_1(\text{size}) + \theta_2(\text{size})^2 + \theta_3(\text{size})^3\end{aligned}$$

$$x_1 = (\text{size})$$

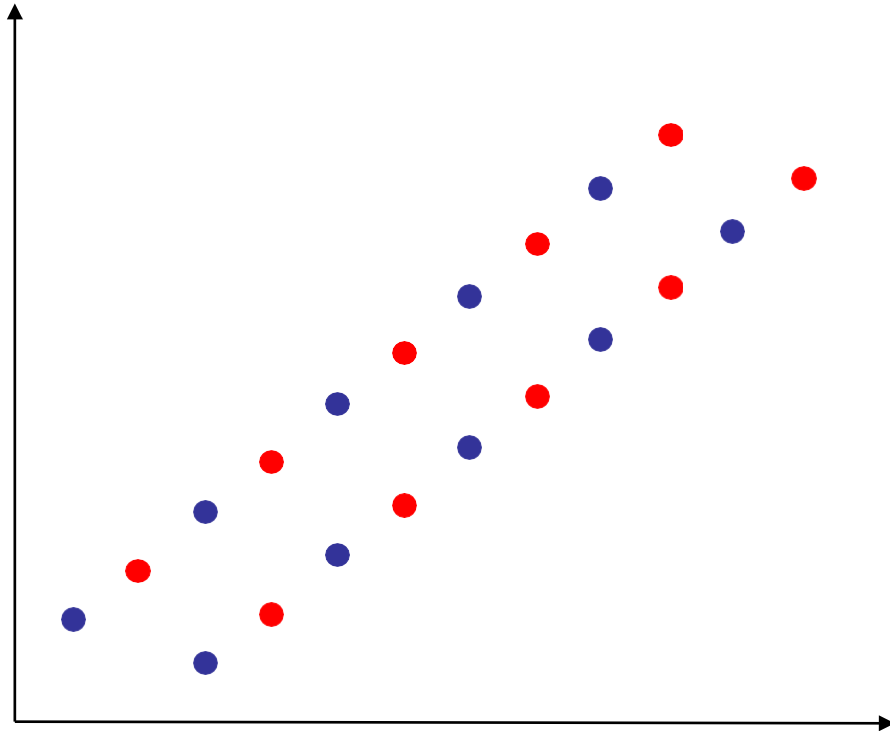
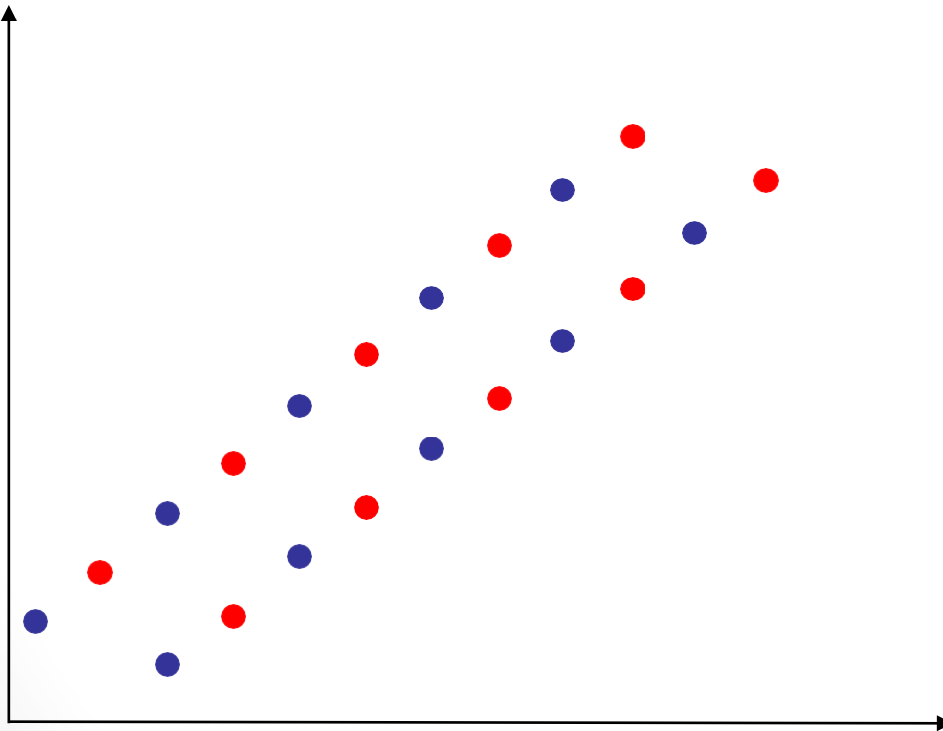
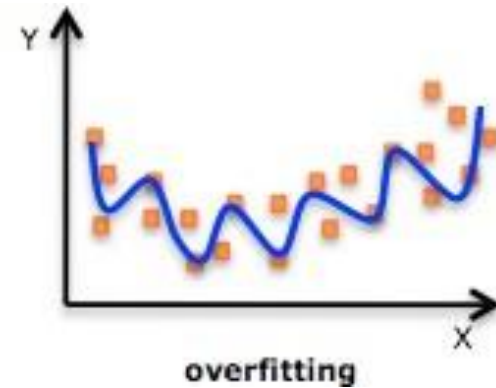
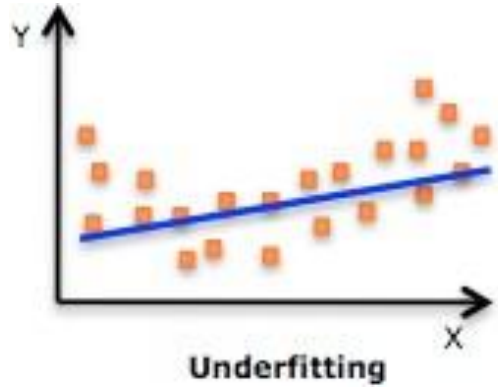
$$x_2 = (\text{size})^2$$

$$x_3 = (\text{size})^3$$

The fitting problem



The fitting problem

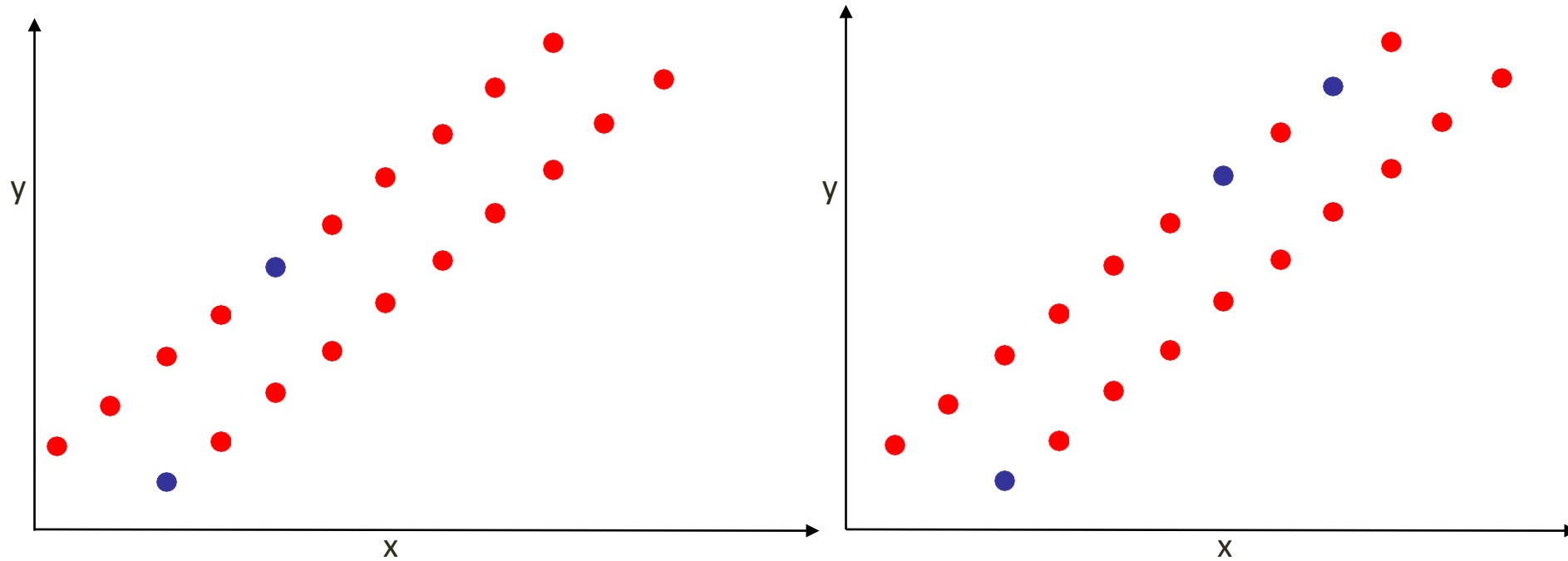


A real example



<http://www.petanimalexpo.co.nz/>

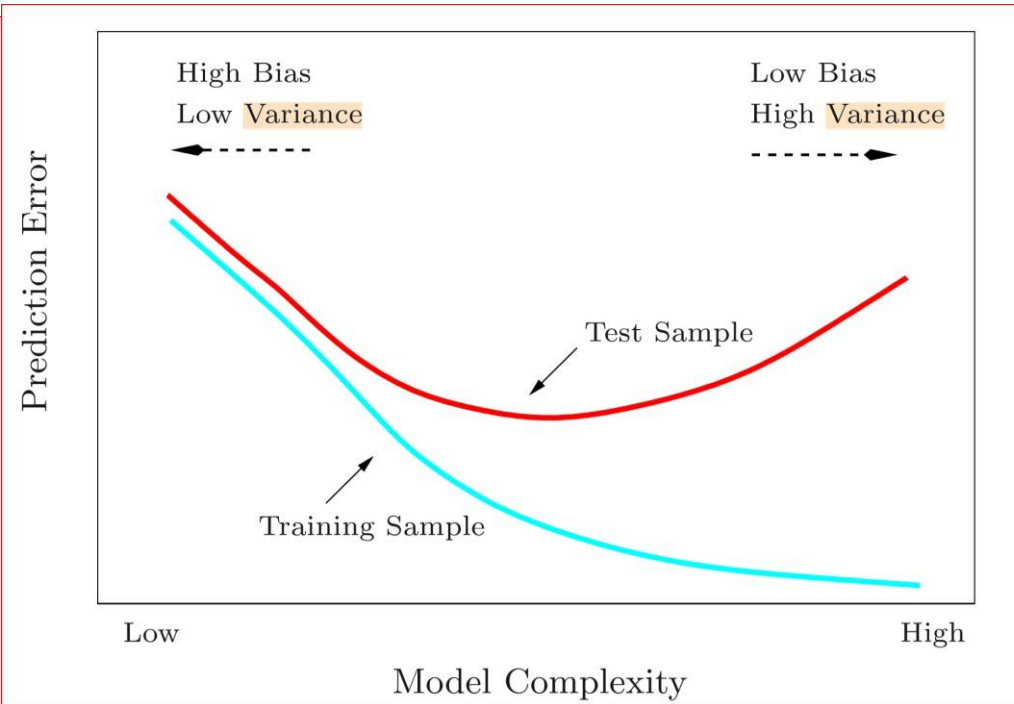
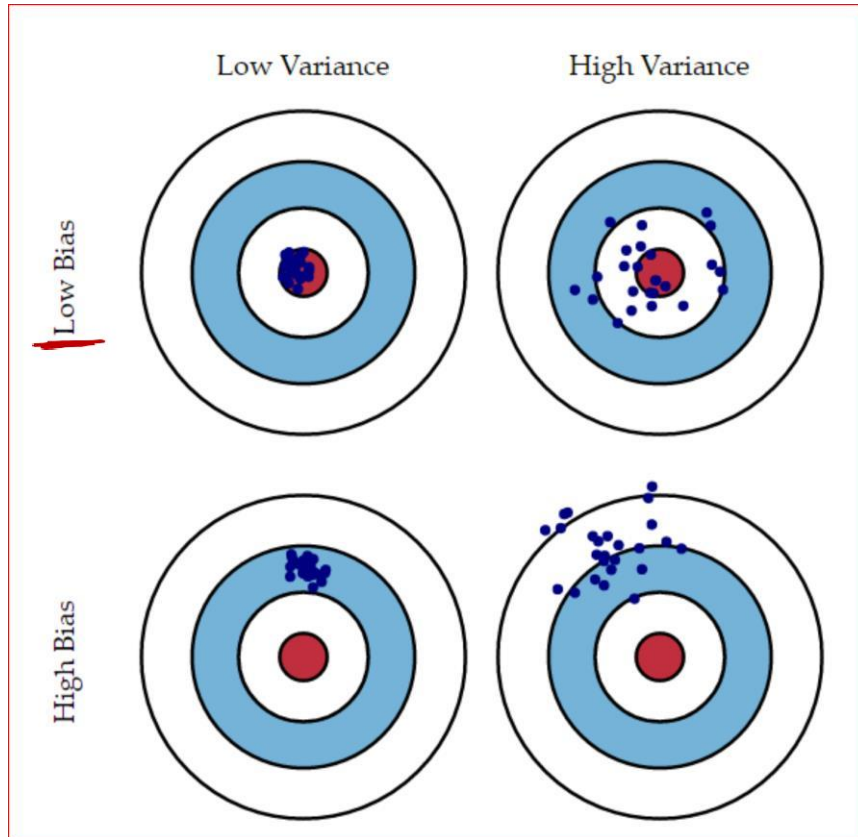
The fitting problem



Bias and Variance

- **Bias** is the difference between the average prediction of our model and the correct value which we are trying to predict.
 - If the average predicted values are far off from the actual values, then the bias is high.
 - Model with high bias pays **little attention to the training data** and oversimplifies (presumes a lot about) the model.
 - High bias causes algorithm to **miss relevant relationship between input and output variable**.
 - When a model has a high bias then it implies that the model is too simple and does not capture the complexity of data thus **underfitting** the data.
 - It leads to high error on **training and test data**.
- **Variance** is the variability of model prediction for a given data point or a value which tells us spread of our data. Variance tells us how scattered are the predicted value from the actual value.
 - Model with high variance pays a lot of attention to training data and does not generalize on the data which it hasn't seen before.
 - As a result, such **models perform very well on training data but has high error rates on test data**.
 - High variance causes **overfitting** that implies that the algorithm models random noise present in the training data.

Bias and Variance



Bias and Variance

- **Is there a way to find when we have a high bias or a high variance?**
- High Bias can be identified when we have
 - High training error
 - Validation error or test error is close to training error
- High Variance can be identified when
 - Low training error
 - High validation error or high test-error
- **How do we fix high bias or high variance in the data set?**
- High bias is due to a simple model and we also see a high training error. To fix that we can do following things:
 - Add more input features
 - Add more complexity by introducing polynomial features
 - Decrease Regularization term
- High variance is due to a model that tries to fit most of the training dataset points and hence gets more complex. To resolve high variance issue we need to work on
 - Getting more training data
 - Reduce input features
 - Increase Regularization term

Solution

- **Reduce the number of features**
 - Manually select features
 - Model selection
- **Regularization**
 - Reduce magnitude/values of parameters θ_j .
 - Works well when we have a lot of features, each of which contributes a bit to predicting .
- **Bagging and Boosting**

Thank You